

Data Warehousing

Master in Data Science

Introduction to Data Warehousing

Knowledge Objectives

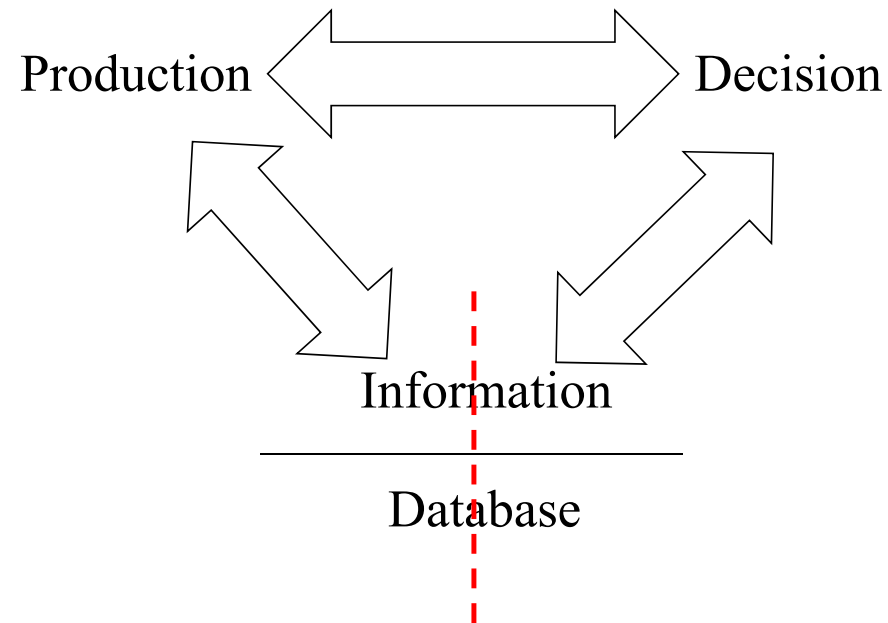
1. Distinguish the three subsystems in every company
2. Distinguish descriptive from predictive analytics
3. Explain the data-driven decision making framework
4. Explain what the “Business Intelligence Cycle” is
5. Identify the different data science flows

Data Driven Decision Making

Subsystems of an enterprise

- Production: Perform activities constituting the goal of the enterprise
- Decision: Plan, Coordinate and Control production activities
- Information: Collect (*input*), Store (*save*), Process (*run*) and Distribute (*output*) all information relevant and needed by the other systems

Kinds of information systems



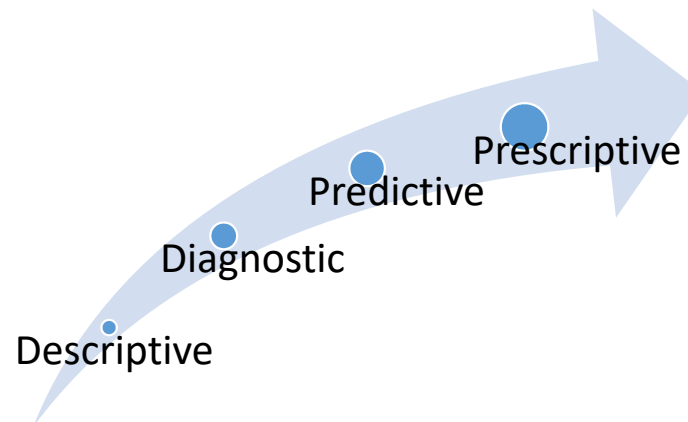
Use of data

"Without data you are just another person with an opinion."

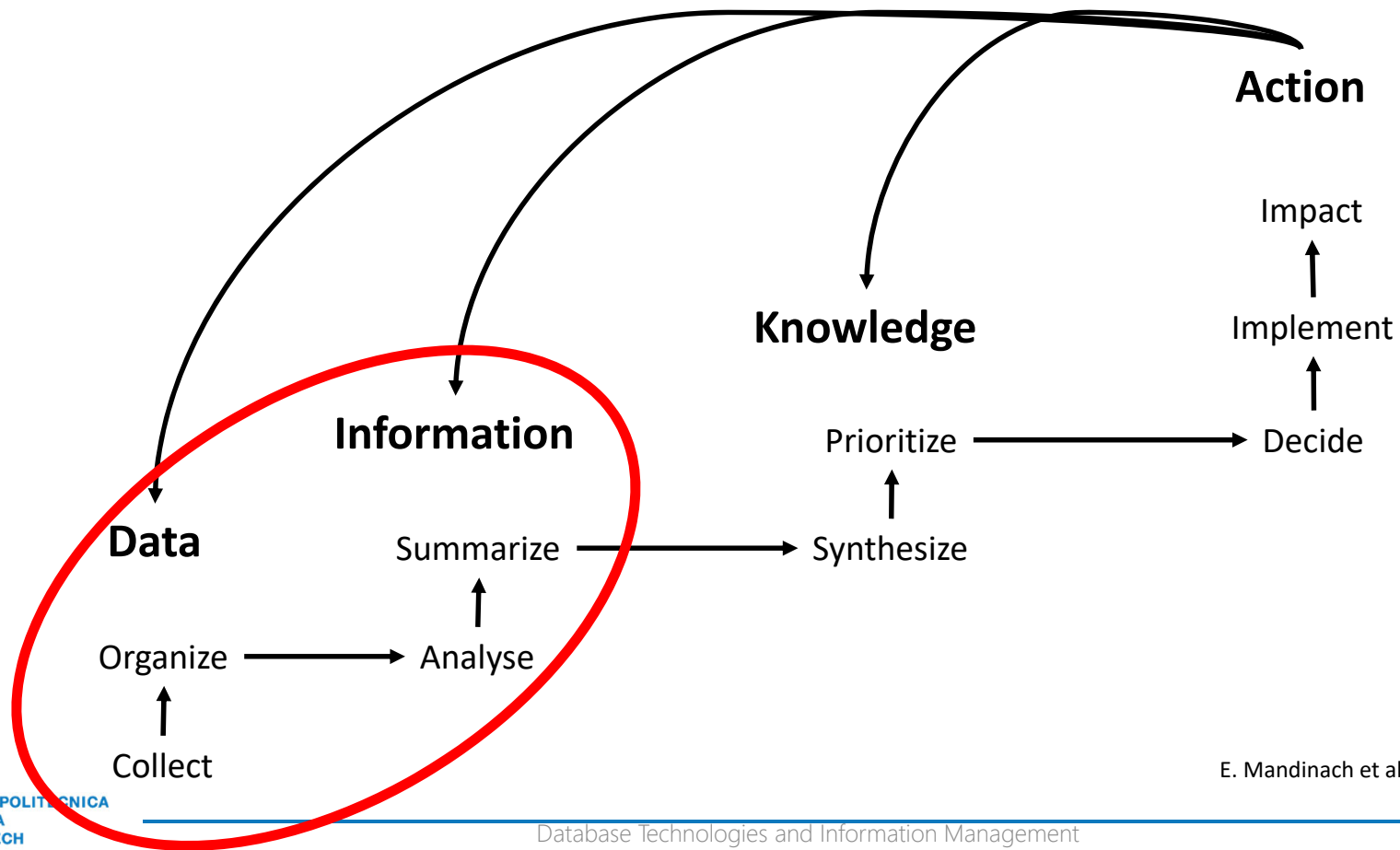
William Edwards Deming

"The world's most valuable resource is no longer oil, but data."

The Economist

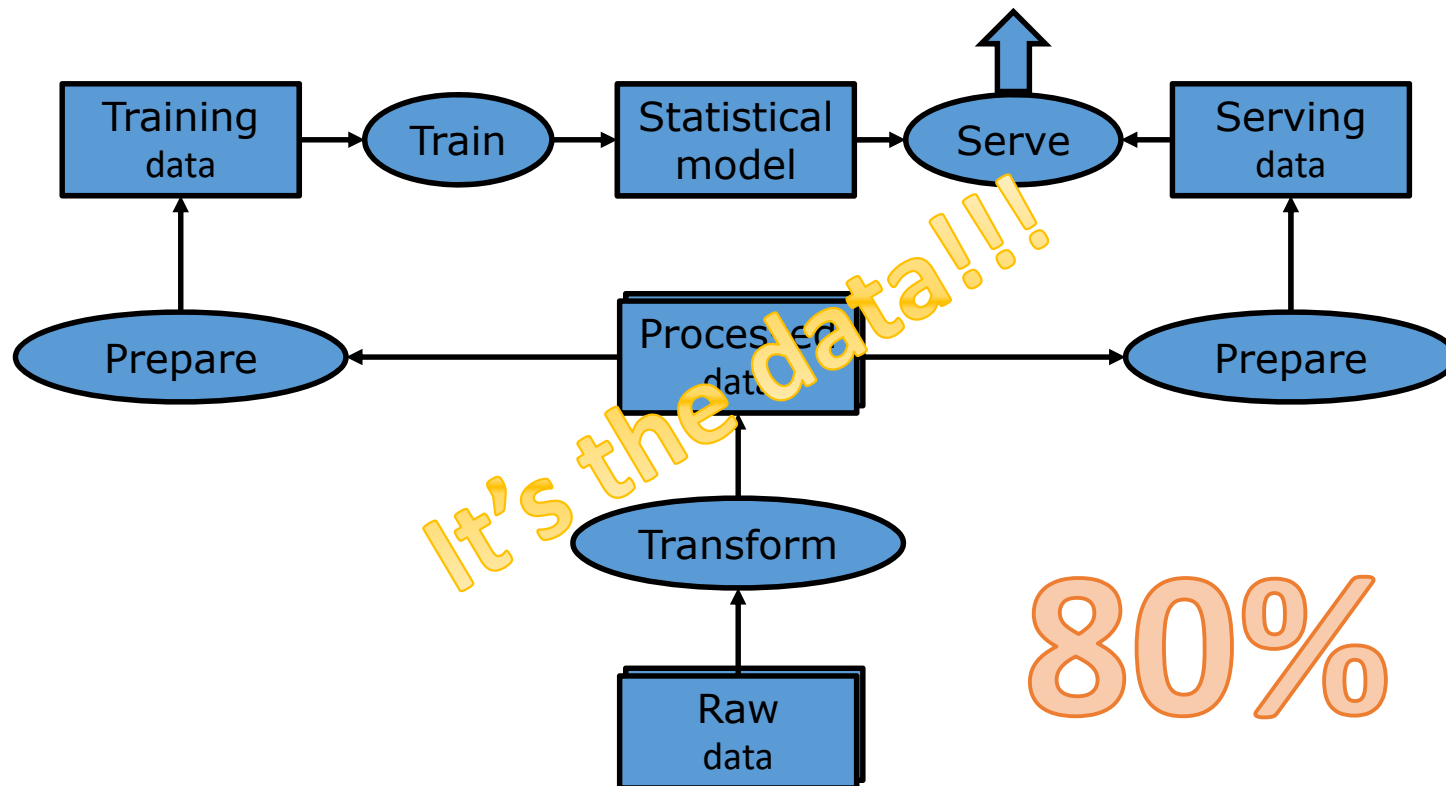


Data driven decision making



E. Mandinach et al.

Data science lifecycle



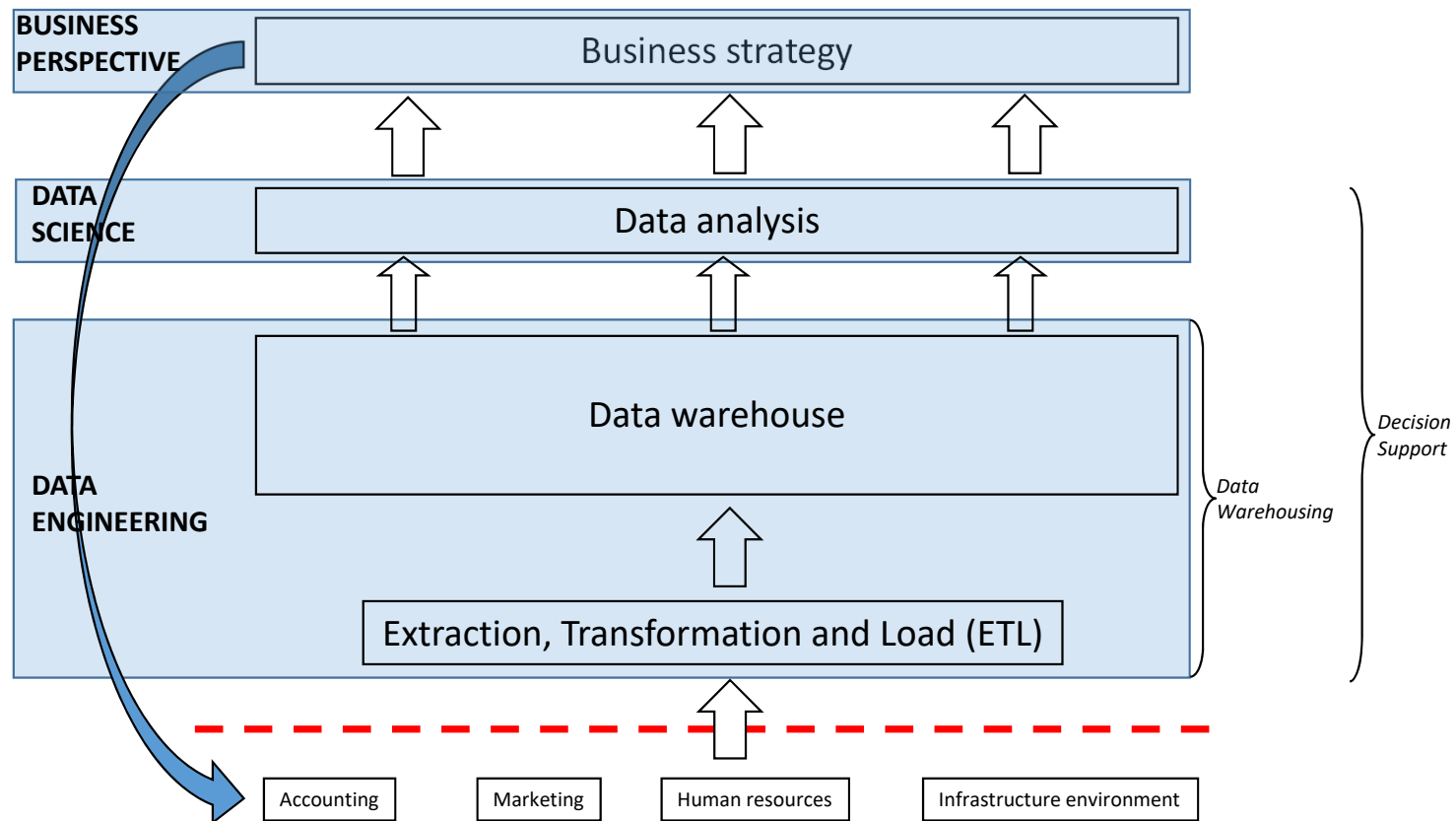
Data-centric AI

- More gains are coming from deeply understanding the data rather than model tweaking
- Custom models are difficult to maintain and extend in a production environment

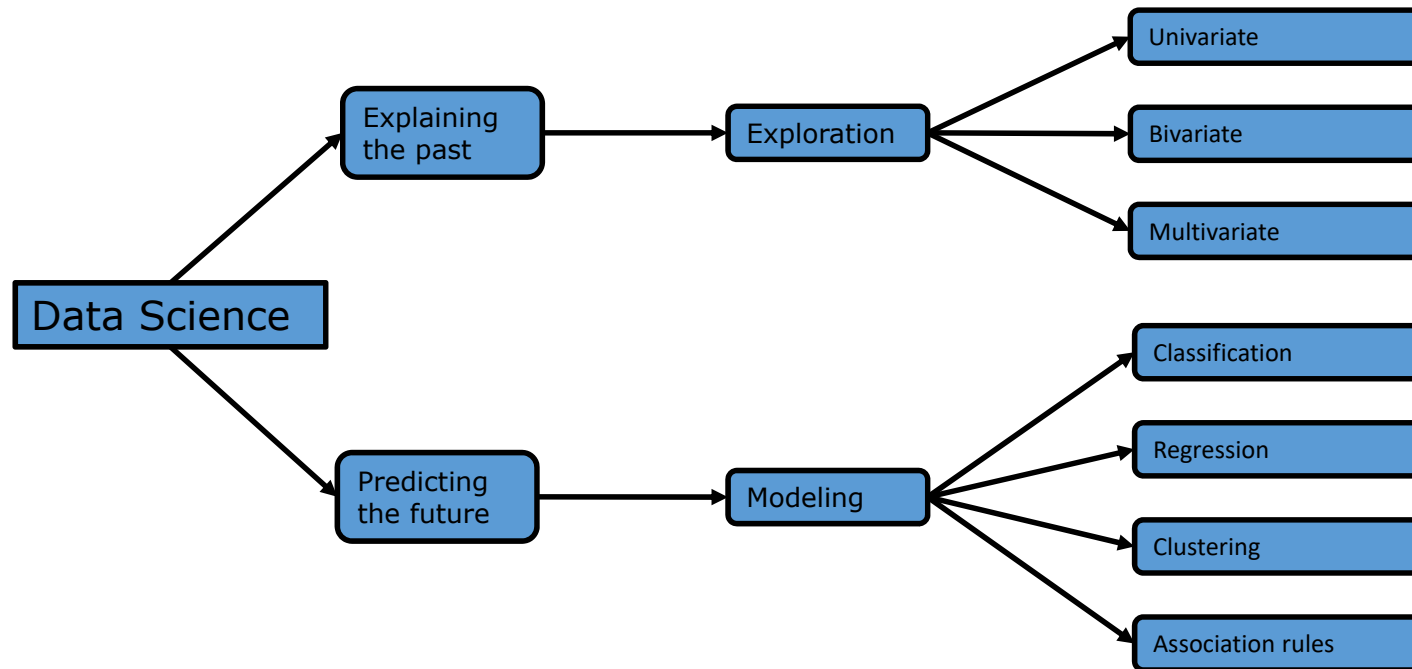
<https://github.com/HazyResearch/data-centric-ai>

Data Engineering and Data Science

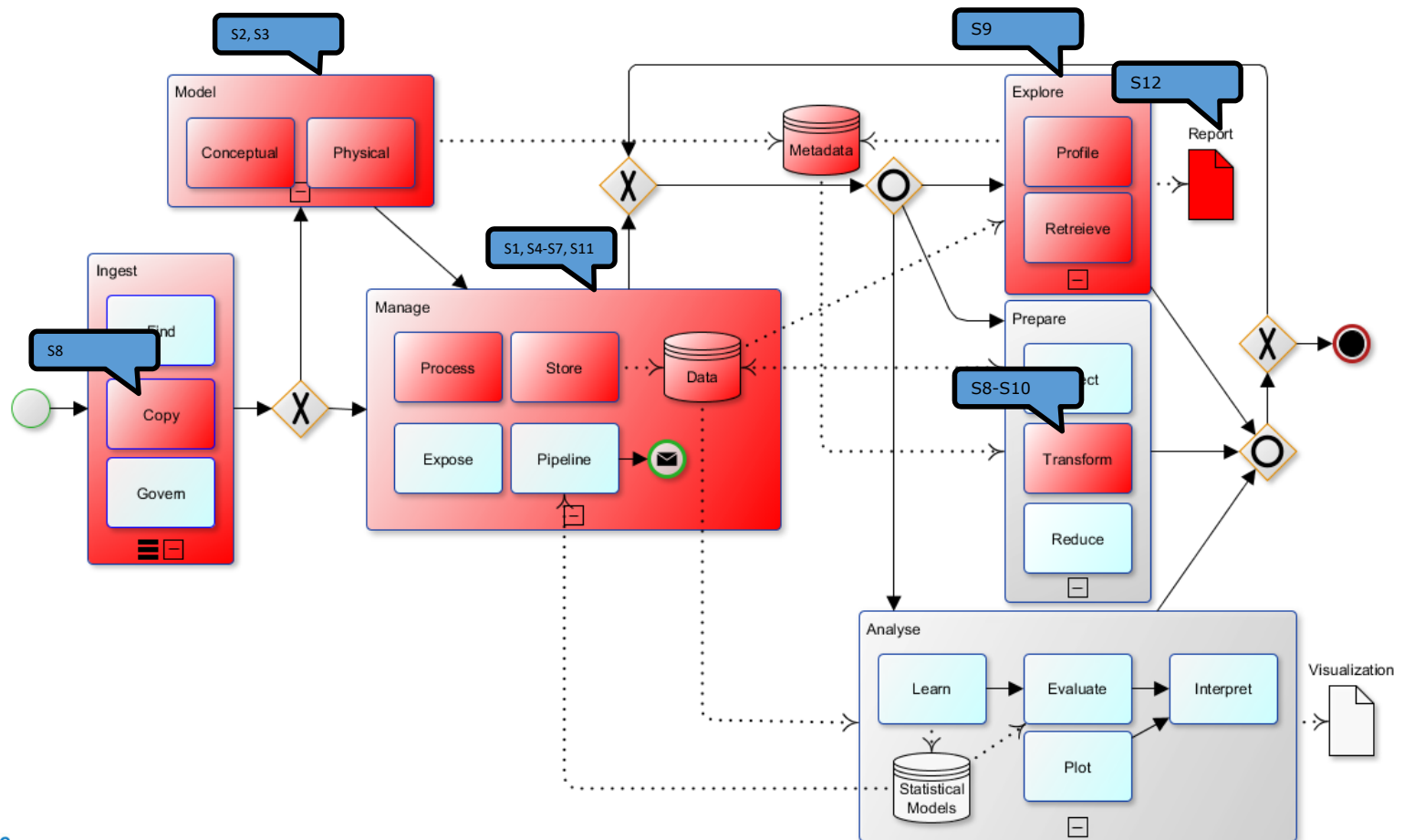
Business Intelligence Cycle



Data Science tools



Data Science Flows

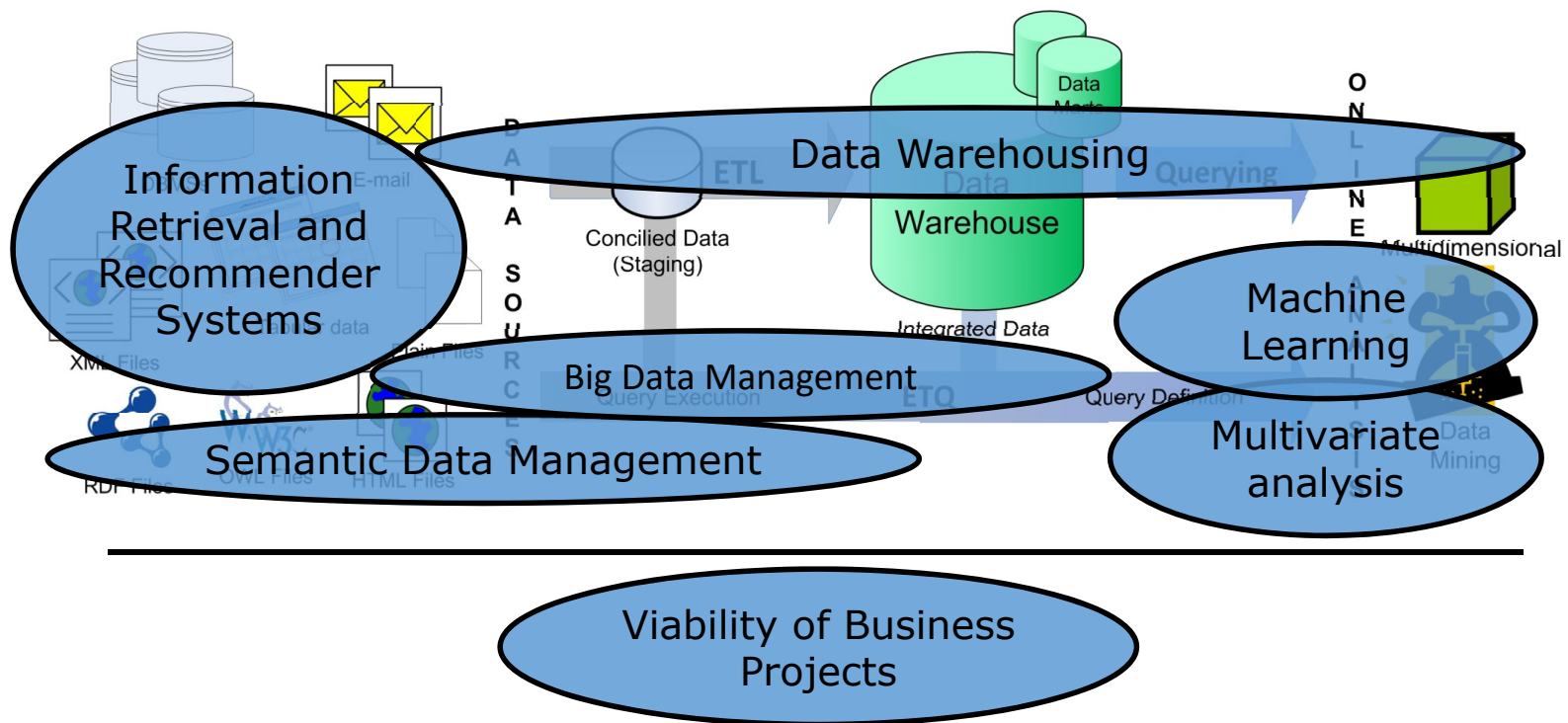


Trends in programming languages

	Platform	2019
	Python	65.8%
	R	46.6%
➡	SQL	32.8%
	Java	12.4%
	Unix shell/awk	7.9%
	C/C++	7.1%
	Javascript	6.8%
	Other programming and data languages	5.7%
	Scala	3.5%
	Julia	1.7%
	Perl	1.3%
	Lisp	0.4%

<https://www.kdnuggets.com/2019/05/poll-top-data-science-machine-learning-platforms.html>

Contextualization



Closing

Summary

- Data-driven decision making
- Descriptive analytics
- Business Intelligence Cycle
- Data science flows

References

- E. Mandinach et al. *A Theoretical Framework for Data-Driven Decision Making*. AERA annual meeting, 2006
- D. Donoho. *50 years of Data Science*.
<http://courses.csail.mit.edu/18.337/2015/docs/50YearsDataScience.pdf>